



Applications of artificial intelligence in rehabilitation: technological innovation and transformation of clinical practice

Haoyang Liu^a, Qiurong Xie^{b,c,*}

^a RainbowFish Rehabilitation&Nursing School, Hangzhou Vocational & Technical College, Zhejiang, China

^b College of Rehabilitation Medicine, Fujian University of Traditional Chinese Medicine, Fujian, China

^c Key Laboratory of Orthopaedics & Traumatology of Traditional Chinese Medicine and Rehabilitation (Fujian University of Traditional Chinese Medicine), Fujian, China

ARTICLE INFO

Keywords:

Artificial intelligence
Personalized medicine
Personalized rehabilitation
AI

ABSTRACT

The integration of artificial intelligence (AI) into rehabilitation science is revolutionizing traditional therapeutic models, offering innovative solutions that enhance the precision, efficiency, and accessibility of rehabilitation services. This review explores the diverse applications of AI in rehabilitation, focusing on key technologies such as machine learning, deep learning, computer vision, natural language processing, and robotics. A key innovation is the proposed AI-empowered rehabilitation model, which transforms fragmented processes into an interactive, adaptive system with real-time assessment during interventions. AI-driven advancements in impairment assessment, intervention planning and delivery, post-discharge care, and patient education are driving a shift from experience-driven to data-model-driven rehabilitation systems. Notable AI-driven applications include AI-powered exoskeletons for motor rehabilitation (e.g., in stroke recovery), NLP-driven cognitive therapy, and tele-rehabilitation platforms that enable remote monitoring and adaptive interventions. Despite these advancements, challenges remain, including data limitations, ethical concerns, regulatory requirements, and clinical integration barriers. Addressing these challenges requires interdisciplinary collaboration to ensure AI's responsible and effective deployment in rehabilitation. This review highlights the transformative potential of AI in rehabilitation and emphasizes the need for continued research and validation to optimize patient outcomes and accessibility.

1. Introduction

Rehabilitation is a pivotal discipline dedicated to enhancing the quality of life for individuals with disabilities or impairments by improving functional abilities and participation in daily activities [1]. The escalating global burden of population ageing, rising prevalence of chronic non-communicable diseases, and the aftermath of health emergencies have intensified the demand for rehabilitation services [2, 3]. The World Health Organization (WHO) has prioritized rehabilitation as a global health strategy, advocating for universal access and integration into health systems, particularly in low- and middle-income countries [4,5]. However, a significant unmet need persists, with up to 50% of individuals who could benefit from rehabilitation services lacking adequate access [6].

The conventional rehabilitation process involves a cyclical sequence of identifying a patient's needs, correlating these with personal and environmental factors, defining rehabilitation goals, planning and implementing interventions, and evaluating outcomes (Fig. 1). These

processes are highly individualized, varying according to patient characteristics. Historically, rehabilitation has relied on personalized care strategies, which are resource-intensive and heavily dependent on clinician expertise and time.

The integration of artificial intelligence (AI) into rehabilitation science is reshaping this landscape by introducing innovative solutions that enhance the precision, efficiency, and accessibility of therapeutic interventions. AI encompasses technologies that enable machines to perform tasks traditionally requiring human intelligence [8], often surpassing human capabilities in data processing and pattern recognition. AI applications in rehabilitation are increasingly recognized for their capacity to transform care delivery. Technologies such as machine learning and neural networks enable the analysis of complex datasets to identify patterns, optimize treatment strategies, deliver real-time assessments, and implement preventive measures. For example, AI facilitates personalized rehabilitation through data-driven robot control and real-time feedback, notably in exoskeleton-assisted rehabilitation for lower-limb impairments [9]. AI-powered platforms deliver

* Corresponding author.

E-mail address: xieqiurong@fjtcu.edu.cn (Q. Xie).

<https://doi.org/10.1016/j.slast.2025.100360>

Received 3 March 2025; Received in revised form 19 October 2025; Accepted 26 October 2025

Available online 27 October 2025

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individualized psychological and nutritional support, enable remote rehabilitation services, and enhance patient engagement through immersive, interactive experiences [10–12]. Recent advancements in generative AI and large language models (LLMs), such as ChatGPT, DeepSeek, and Claude, have demonstrated significant potential in personalized therapy design, clinical education, and patient interaction [13,14]. Moreover, AI-driven movement analysis and injury risk prediction enable proactive preventive strategies [15], aligning with rehabilitation’s dual goals of treating existing conditions and preventing future impairments. Recent data highlight the rapid adoption of AI, with healthcare data volumes projected to grow at a 36 % compound annual growth rate by 2025 [16].

Technical pathways, including continuous data streaming for real-time feedback, AI algorithms for dynamic intervention adjustments, and predictive modeling for adaptive goal-setting, facilitate the transition from experience-driven to data-driven rehabilitation. This study proposes a novel AI-empowered rehabilitation model (Fig. 2) that integrates these elements into a cohesive, interactive cycle, enabling unique applications such as autonomous exoskeleton adjustments and LLM-driven cognitive simulations in home-based settings. However, challenges remain, including limited generalizability of AI models due to data constraints[17], insufficient applications tailored to specific populations, such as pediatric rehabilitation [18], and ethical concerns related to data privacy and algorithmic bias [10].

This paper explores the diverse applications of AI in rehabilitation science, highlighting its potential to revolutionize traditional models. It further examines critical challenges, including data limitations, ethical considerations, regulatory complexities, and barriers to clinical integration, which must be addressed to realize AI’s benefits fully. By overcoming these obstacles, AI can significantly enhance the accessibility and efficacy of rehabilitation services for diverse populations.

This review distinguishes itself from existing literature on AI in rehabilitation by proposing a novel AI-empowered rehabilitation model that integrates real-time adaptive cycles. Unlike the descriptive overviews common in prior works, this model often lacks conceptual innovation. The review incorporates critiques with PRISMA-guided tables, including effect sizes and quality assessments instead.

2. Brief in AI technologies

The integration of AI into rehabilitation relies on a suite of advanced technologies that enable data-driven decision-making, personalized interventions, and real-time monitoring. Key AI technologies include machine learning (ML), deep learning (DL), computer vision (CV), natural language processing (NLP), and robotics. These technologies overlap in their applications, with DL being a subset of ML that leverages neural networks for automated feature extraction, while CV and NLP often utilize DL techniques for specific tasks. This section explains their features, specific algorithms, and typical use cases in rehabilitation, supported by real-world implementations.

Machine Learning involves algorithms that learn patterns from data to make predictions or decisions without explicit programming, enabling adaptive and personalized solutions in fields like rehabilitation. Supervised learning models, such as support vector machines (SVMs) and decision tree models, train on labelled data (e.g., patient outcomes paired with clinical metrics) to predict future events. They are widely used for predicting functional recovery outcomes by analyzing clinical, sensor, or self-reported health data [19,20]. Unsupervised learning techniques, like clustering, identify hidden patterns in unlabeled data. For example, stratifying patients into subgroups based on mobility or cognitive profiles enables tailored therapy plans [21].

Reinforcement learning (RL) uses trial-and-error feedback to optimize actions in dynamic environments (e.g., adjusting robotic assistance levels) [22]. Specific algorithms, such as Deep Deterministic Policy Gradient (DDPG), have been applied in exoskeletons combined with sliding mode control for upper-limb exoskeletons. The experimental data show an improvement of 11.89 % compared to the computed torque method [23].

Deep Learning (DL), a subset of ML, uses multi-layered neural networks to process complex, high-dimensional data, automating feature extraction. Convolutional Neural Networks (CNNs) apply filters to detect spatial patterns, such as in video-based gait analysis, achieving 90 % accuracy in detecting abnormalities in real-time applications [24]. Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory (LSTM) variants, process sequential data from IoT devices,

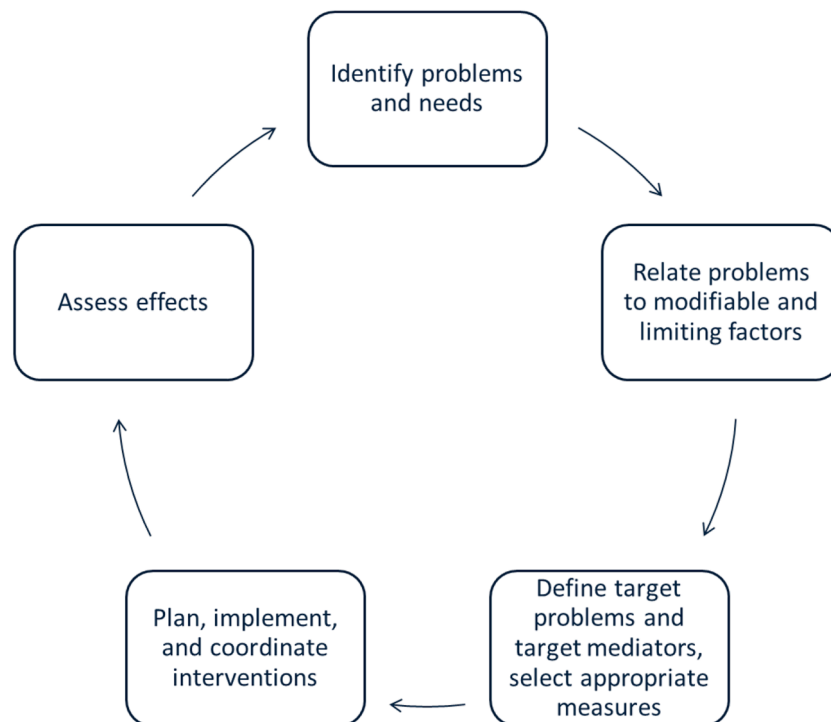


Fig. 1. The conventional rehabilitation cycle adapted from [7].

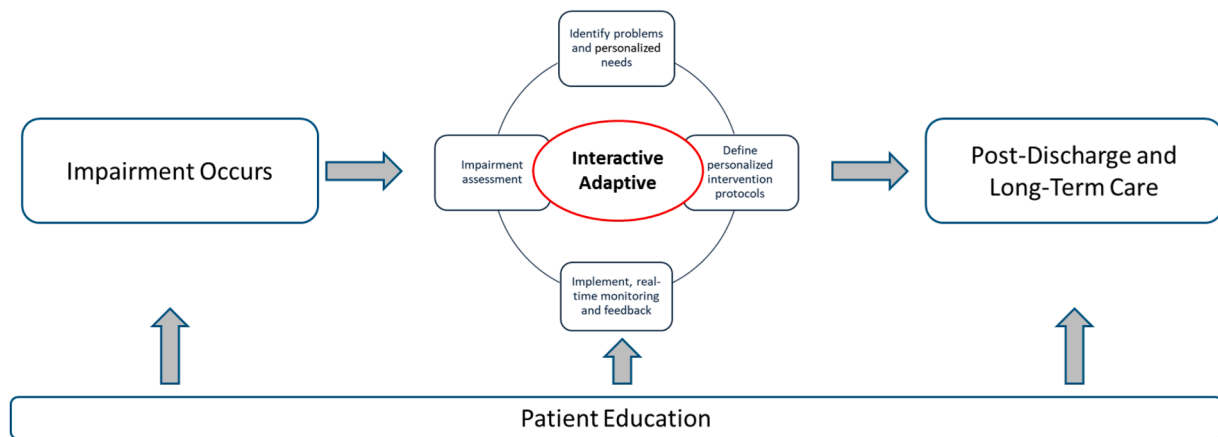


Fig. 2. New rehabilitation model empowered by AI technologies.

improving rehabilitation efficiency, recovery speed, and reducing re-injury risk [25,26]. The overlap between ML and DL lies in their shared goal of pattern recognition, but DL's neural architecture enables it to handle raw, unstructured data more effectively.

Computer Vision (CV) enables machines to interpret visual data (e. g., images, videos). Markerless motion capture systems, powered by algorithms like OpenPose, track patient movements during therapy, quantifying joint angles and gait symmetry with 95 % precision in Parkinson's disease assessments [27]. These systems reduce manual assessment burdens and provide clinicians with objective metrics for rehabilitation progress.

Natural Language Processing (NLP) supports cognitive and communication rehabilitation by analyzing and generating human language. Transformer-based Large Language Models (LLMs), such as GPT-4, analyze speech patterns in aphasia patients, achieving 80 % accuracy in tracking language recovery [13,14]. LLMs also generate personalized patient education content, improving adherence in stroke rehabilitation programs [14,28].

AI-Driven Robotics and Exoskeletons merge with ML with mechanical systems to deliver adaptive physical therapy. Robotic devices, such as upper-limb exoskeletons, use sensor fusion and adaptive control algorithms to adjust assistance levels based on real-time patient performance [29]. Similarly, brain-computer interfaces (BCIs) employ AI to decode neural signals, enabling paralyzed patients to control robotic limbs or communication devices [30].

Other emerging technologies include digital twins—virtual replicas of patients—that simulate rehabilitation scenarios to optimize treatment plans [31]. For example, digital twins have been implemented in post-stroke rehabilitation to model patient-specific biomechanical responses, enabling personalized rehabilitation plans [32]. Federated learning frameworks represent a decentralized ML approach where models are trained across multiple institutions without sharing raw data, addressing data privacy concerns while maintaining compliance with regulations like GDPR [33]. In rehabilitation, Studies show FL-trained models outperform traditional isolated models by 20–30 % in robotic rehabilitation tasks and achieve similar or better accuracy compared to centralized approaches, all while maintaining privacy[34,35].

These technologies collectively drive innovation in rehabilitation, with clear delineations (e.g., DL's reliance on ML frameworks) and synergies (e.g., CV and NLP leveraging DL algorithms). Their clinical effectiveness depends on robust implementation and validation, as discussed in subsequent sections.

3. New rehabilitation model and enabling AI applications

Rehabilitation is a cyclical, multi-stage process involving assessment, intervention planning, and delivery, as depicted in Fig. 1. Conventional

rehabilitation often relies on standardized protocols that do not adjust dynamically to individual progress or needs, resulting in under- or over-challenging therapy [36–38]. This rigidity can reduce engagement and limit recovery potential. The fragmented rehabilitation process causes poor coordination between hospital and community care, unclear care plans, and insufficient communication among providers. Such process leads to delays, unmet needs, and feelings of abandonment among patients and caregivers, especially during transitions from inpatient to community settings[39–41].

AI technologies are transforming this process by integrating real-time feedback and adaptive interventions, addressing gaps in traditional models that lack dynamic personalization and scalability. This section proposes a novel AI-empowered rehabilitation model (Fig. 2), derived from identified literature gaps, such as limited real-time adaptivity and fragmented care processes [17,18]. The model transforms previously disconnected stages: assessment, planning, implementation, and adjustments—into an interactive, adaptive system. This integration enables simultaneous assessment during interventions, with AI dynamically adjusting protocols to optimize outcomes, such as through reinforcement learning (RL) for robotic assistance or large language models (LLMs) for patient education. Advancements in AI and digitalization also democratize medical knowledge, making it more accessible and understandable, thus encouraging proactive patient behaviours.

3.1. Impairment assessment

AI expands rehabilitation assessment beyond clinical settings into home environments, leveraging non-invasive technologies to enhance precision and accessibility. Traditional methods often rely on subjective clinician observations or labour-intensive manual analyses. AI-driven tools automate and refine these processes, enabling earlier detection of impairments and data-driven decision-making. In this subsection, we discussed two significant categories of rehabilitation needs: motor and cognitive impairments. A summary of the investigated literature, including sample size, effect sizes, and quality assessments (informed by PRISMA guidelines for systematic reviews), is listed in Table 1. These studies demonstrate a trend toward robust, interpretable AI models (e. g., ensemble methods, XAI) with increasing sample sizes and external validation. However, many still face limitations in sample size, heterogeneity, and standardization of effect metrics. Most reviews and primary studies adhere to PRISMA or similar systematic review standards, with explicit reporting of inclusion/exclusion, validation, and bias assessment.

3.1.1. Automated motor function assessment

AI-powered non-invasive systems, including computer vision (CV), deep learning (DL), machine learning (ML), reinforcement learning

Table 1

Summary table on AI-based functionality assessment in rehabilitation, including design, metrics, and PRISMA-informed quality.

Study & Population	AI Technology Used	Sample Size	Effect Size	Quality Assessment
[42] Tele-rehabilitation	CV	$n = 14$ (5 healthy, 9 stroke)	Mean error: 1.75 %; good match with VICON	Small samples; technical validation; limited clinical scope
[43] Motor rehabilitation	CV/DL	REHAB24-6 dataset (~2500 images/videos); single case study (volleyball player with rotator cuff injury)	Model accuracy: 99.29 % (train), 99.88 % (test); latencies <10–120 ms	Prototype stage; simulation-only testing; single-user; needs clinical validation
[44] Unresponsive brain injury	CV	$n = 53$ (37 ABI patients, 16 healthy subjects); 423 analyzable videos	SeeMe detected responses in 30/36 patients (eyes), 16/17 (mouth); more sensitive than raters (5.4 vs 2.8 positive trials/10); AUC-ROC: 0.70 for outcome prediction; classifier accuracy: 65 %	Single-center; small subsample; needs multi-center validation; IRB-approved, ClinicalTrials.gov registered; high-quality prospective design
[45] Parkinson's disease	CV	729 videos (544 PD + 185 healthy)	Sensitivity: 73 %, Specificity ≈ 68 %; high correlation (r , MSE)	Large sample; strong diagnostic validation; not RCT
[46] Post-stroke rehab	ML (linear/logistic regression, SVM, cross-validation)	19 studies (varied n)	RMSE 0.75–22.6 %	Small samples; limited external validation; high heterogeneity
[47] Sub-acute stroke	Random Forest + Explainable AI (XAI)	$n = 95$	ML outperformed stats; feature relevance stable; accuracy not numerically specified	Robust design, interpretable ML, clinical relevance
[48] Spinal cord injury	CatBoost (Tree-based ensemble ML)	$n = 3827$	AUC 0.85, accuracy 0.77, R^2 0.77, MAE 9.5	Large sample, external validation, transparent reporting
[49] Neuromotor disorders	SVM, Tree-based ensemble ML	$n = 46$	F1 score: 95.6 % (self), 95.4 % (therapist)	Nested k-fold CV, detailed reporting, small sample
[50] Ortho/neuro rehab	Tree-based ensemble ML	$n \sim 4050$	RMSE: 6.6 (ortho), 8.7 (neuro)	Robust preprocessing; needs more external validation
[51] Hip fracture rehab	SVM, ensemble ML	$n = 1625$	ML > regression; AUC not specified	Large sample; model comparison; clear predictors
[52] Post-stroke upper limb	Multi-modal ML	$n = 30$ (15 patients, 15 controls)	Accuracy up to 96.1 %, $R = -0.87$ with clinical scores	Small sample; strong correlation with clinical gold standard
[53] Stroke rehab	RL	$n = 22$ (15 patients, 7 therapists)	F1-score improved from 0.66 to 0.71 ($p < 0.01$)	Human-centered; usability focus, small sample
[54] Post-stroke home rehab	Regression/classification ML	$n = 20$	ADL accuracy 94.4 %, RMA up to 91.3 %	Small sample; personalized approach; clear metrics
[55] Alzheimer's disease	CV	3714 MRI Brain scans: NonDemented (834 images), MildDemented (1824 images), and VeryDemented (1056 images).	Accuracy: 99.32 % (self) 86.6 % and 99.5 % (external)	Large sample; external validation
[56] Hand performance	Wearables/ML	$n = 7$ (4 healthy, 3 stroke patients)	Accuracy: 85 % (± 5.1 %);	Small sample size; sensor inaccuracies for varying hand sizes; not suitable for low-recovery patients
[57] Stroke	Wearables/DL	$n = 48$ (chronic stroke individuals)	Classification accuracy: 70 %	Reliance on human labelling; potential distributional shifts
[58] Aphasia	NLP	$n = 109$ (92 aphasia patients, 17 controls)	Correlation of 0.827	Language-specific; small sample; no longitudinal data
[59] ADHD	Eye-tracking/AI	$n = 437$ (112 ADHD, 325 TD children)	diagnosis and age had significant effects on performance on the fixation task, $p < 0.001$	Short task duration; minimum trials may be insufficient; potential biases
[60] Dementia	Eye-tracking/ML/VR	$n = 201$ (elderly participants grouped by education)	AUC: 0.88–0.94; sensitivity: 88.5 %; specificity: 83 %; $r = 0.90$	Single-city sample; no comorbidity data; potential visual impairment biases

(RL), and wearables, quantify motor performance with high accuracy. CV and DL assess gait and movement in tele-rehabilitation and motor disorders (e.g., stroke, Parkinson's), with sample sizes from 14 to 729, achieving metrics like 1.75 % mean error, 73 % sensitivity, and 99.29–99.88 % model accuracy, though limited by small samples or prototype stages in some cases [51,52,54,57]. ML techniques, including SVM, tree-based ensembles (e.g., CatBoost), and multi-modal ML, evaluate neuromotor disorders, spinal cord injury, hip fracture, and post-stroke upper limb recovery, with sample sizes from 15 to 4050, yielding AUC 0.85, F1-scores up to 95.6 %, and RMSE 6.6–8.7, despite challenges like small samples and limited validation [42,44,45,48–50]. RL enhances stroke rehabilitation usability (15 patients, 7 therapists), improving F1-scores to 0.71 [46]. Wearables with DL assess chronic stroke (48 patients, 70 % accuracy), constrained by human labelling [57]. These tools reduce clinical visit dependence, supporting rural access and real-time feedback in the novel model.

3.1.2. Cognitive and communication disorder evaluation

AI advances cognitive and communication assessments through ML,

random forest (RF) with explainable AI (XAI), CV, natural language processing (NLP), and eye-tracking, targeting disorders like aphasia, Alzheimer's, ADHD, and dementia. ML and RF/XAI assess post-stroke and sub-acute stroke recovery (19 studies, 95 patients), showing RMSE values of 0.75–22.6 % and stable feature relevance, though limited by heterogeneity [42,43]. CV analyzes 3714 Alzheimer's MRI scans, achieving 99.32 % self-accuracy and 86.6–99.5 % external validation [53]. NLP evaluates aphasia in 92 patients, with a correlation of 0.827, constrained by language specificity [58]. Eye-tracking with ML/VR assesses ADHD (437 children) and dementia (201 elderly), showing an AUC of 0.88–0.94 and significant task performance effects ($p < 0.001$), though limited by short task durations or single-city samples [59,60]. These tools standardize assessments, reduce subjectivity, and align with the novel model's dynamic, adaptive framework for continuous monitoring.

From the existing literature, AI-assisted assessment applications have expanded the diagnosis and assessment methods from the medical environment to the home or mobile environment. Diagnosis or assessment is no longer an independent operation in the traditional

rehabilitation cycle; instead, it can be dynamically integrated into rehabilitation interventions.

3.2. Intervention planning and delivery

The integration of AI into rehabilitation intervention planning and delivery has shifted the paradigm from generalized to personalized strategies[61]. Conventional prescribed interventions in rehabilitation involve a stable protocol within a specific cycle. Adjustments to therapeutic interventions occur at the start of the next cycle. AI technologies, including machine learning (ML), deep learning (DL), reinforcement learning (RL), and multi-modal systems, optimize treatment plans and interventions for conditions like stroke, neuromotor disorders, and sensory impairments. Explainable AI (XAI) enhances decision-support systems (DSS), providing interpretable recommendations that allow clinicians to refine plans dynamically. Table 2 summarizes key applications. Considering that a large number of studies in the subsection provide descriptive work on framework feasibility, we selected studies with clearly reported quantitative indicators listed in Table 3.

By leveraging ML, digital twins, and collaborative frameworks, AI enables clinicians to design targeted therapies that maximize recovery potential while minimizing costs and inefficiencies. AI is the key to precision medicine, serving human personalized needs. We synthesize key applications in this domain, drawing on recent research.

Personalized intervention planning involves predicting patients' recovery. An AI-powered Decision Support System (DSS) used in hospitals can predict a patient's future recovery based on their medical profile, thereby guiding early prognosis and specifying personalized rehabilitation strategies [47,48,63,64]. Such applications commonly use interpretable decision tree models, emphasizing that doctors can adjust the corresponding rehabilitation plans based on the suggestions provided by AI.

Wearable devices are suggested for personalized rehabilitation as well [62]. Wearable devices provide real-time analysis and feedback capabilities and can provide targeted guidance to patients. Exoskeletons, as a type of wearable device, can not only provide monitoring and analysis capabilities, but also intervene in patients' movements in real time. Exoskeletons offer mobility assistance and promote neuroplasticity in individuals with impaired motor functions through sensor integration, data analysis, AI-enhanced control systems, and AI-engaged autonomous functionality [9]. The incorporation of AI-powered sensors enhances the functionality of exoskeletons by collecting and analyzing physiological signals such as electromyography (EMG), electroencephalography (EEG), and kinematic data [65–67]. AI-driven analytics processes these signals to refine motor assistance, detect anomalies, and provide real-time feedback to therapists and patients. Such integration helps in improving motor learning and neuro-rehabilitation strategies. AI-driven control mechanisms, such as adaptive impedance control and machine learning-based gait pattern recognition, have enabled exoskeletons to provide precise and dynamic assistance [68–70]. These systems analyze real-time biomechanical data to adjust support levels based on user movement patterns, reducing strain and optimizing rehabilitation outcomes. AI algorithms, typically

DL and RL algorithms, can also predict movement intentions, allowing for smoother and more natural user interaction with the exoskeleton[66, 67,71]. AI enables exoskeletons to operate in semi-autonomous or fully autonomous modes, reducing reliance on external supervision. Through predictive analytics and real-time adjustments, exoskeletons can guide users through rehabilitation exercises with minimal intervention from therapists [72–74]. This not only enhances patient independence but also extends access to rehabilitation for those in remote or underserved areas.

AI is playing a pivotal role in cognitive and communication rehabilitation by providing intelligent support for individuals with speech, language, and cognitive impairments. AI-driven tools are being used to personalize therapy, improve communication strategies, and enhance cognitive functions in patients with conditions such as stroke, traumatic brain injury, and neurodegenerative diseases. AI-powered applications, such as NLP and speech recognition, have been integrated into rehabilitation programs to assist individuals with speech and language disorders. These technologies help assess speech patterns, provide real-time feedback, and adapt therapy exercises to the patient's progress [75–77]. Additionally, AI-driven speech synthesis enables alternative communication methods for individuals with severe speech impairments. Augmentative and alternative communication (AAC) devices now integrate AI-driven predictive text, emotion recognition, and gesture-based interfaces, making them more intuitive and user-friendly. VR applications provide interactive conversational environments that help improve communication skills. AI-driven avatars can simulate human interactions, allowing patients to practice speech and language exercises in realistic social settings. These enhancements improve communication for individuals with motor and speech impairments, increasing their independence and social engagement [78,79]. Machine learning algorithms combined with game or VR techniques assess cognitive performance and tailor training programs to improve memory, attention, and problem-solving skills [80,81]. AI-driven neurofeedback systems monitor brain activity and provide interactive cognitive exercises that enhance neuroplasticity and rehabilitation outcomes [82].

As summarized in Tables 2 and 3, dynamic adaptation and personalization become the major features during intervention planning and delivery. Various tools from physical (e.g., exoskeletons) and virtual (e.g., VR) provide a patient-centred experience. These technologies enhance the precision of intervention exercises, thereby improving the outcome of rehabilitation.

3.3. Post-discharge and long-term care

AI reforms the model of post-discharge and long-term rehabilitation care by enabling continuous monitoring, personalized therapy, and remote patient engagement. These AI-driven applications support individuals in maintaining functional improvements and preventing relapses.

Wearable sensors and AI-driven health applications provide real-time monitoring of patients' physiological and functional data post-discharge. These systems detect anomalies, track recovery progress, and alert healthcare providers in case of potential complications,

Table 2

Summary of investigated studies in rehabilitation intervening planning and delivery. DL - deep learning, RL - reinforcement learning, NLP - natural language processing, VR - virtual reality.

Type of impairment	Tasks	Enabling technologies	Features	Reference
Motor Function	Personalized intervention planning	Patient medical profiles and Tree model	Explainable AI	[11,47,48,63]
	Exoskeleton - patient intention recognition	Robots, DL, and RL	Real-time feedback	[65–67,71]
	Exoskeleton - dynamic assistance	Robots, DL, and RL	Dynamic, precision, automated assistance	[68–70, 72–74]
Cognitive and Communication Disorder	Communication training	Audio analysis, NLP, VR, and DL	Adaptive therapy exercises	[75–80]
	Cognitive training	VR, Supervised learning, and DL	Adaptive therapy exercises	[80–82]

Table 3

Summary of selected AI-driven rehabilitation intervention studies, including design, effect size, and quality.

Study & Population	AI Technology Used	Sample Size	Effect Size	Quality Assessment
[83] Chronic Stroke	Random Forest, KNN, ANN, SVM, Logistic Regression	132	RF: 85 % accuracy (AUC-ROC: 0.86); KNN: 82.5 % (AUC-ROC: 0.76)	Clear inclusion/exclusion, cross-validation; robust reporting
[84] Chronic Stroke	KNN, ANN	239	KNN: 85.4 % accuracy (AUC-ROC: 0.89); ANN: 81.3 % (AUC-ROC: 0.77)	Training/test split; cross-validation, transparent predictors
[85] Stroke	Multi-modal HMI (EMG+FMG), AI decoding	Not specified	91.3 % gesture decoding accuracy	Device validation; robust metrics; clinical application
[86] Home-based	Hybrid ML (XGBoost, KNN, stacking)	$n = 8$	Up to 100 % accuracy (FTSTS/TUG); 5–15 % improvement over prior	Benchmarking, simulation + real data; ART compliance
[87] Upper limb	ML on wearable sensor data	$n = 37$ (16 stroke survivors and 21 TBI survivors)	R^2 : 0.86 (impairment), 0.79 (movement quality)	Clinician comparison; robust stats; real-world data
[71] Motion intention	DL	$n = 101$ (MotionID dataset); n =unspecified for HumanAct12	Accuracy improvements: 9.81 % over ELM; sensitivity up to 8.47 %	Not empirical; no quantitative effects

ensuring timely interventions[88]. The remote assessments of functionalities as mentioned in Section 3.1 are the enabling technologies of tele-rehabilitation [56,89]. AI-based predictive models also help identify patients at risk of long-term mortality after conditions such as stroke, enabling better discharge planning and rehabilitation support [90]. AI-driven virtual assistants provide reminders for medication, therapy exercises, and lifestyle modifications. These systems offer psychological support, helping patients stay engaged with their rehabilitation programs while improving adherence to long-term treatment plans [91]. AI-based cognitive rehabilitation interventions further aid in post-discharge care by delivering personalized exercises for cognitive recovery [92]. Critiques of these applications include limited generalizability due to small, homogeneous samples (e.g., $n = 50$ in [88]) and potential algorithmic biases.

3.4. Patient education

Patient education aims to enhance patient awareness and prevent possible injuries. Compared to traditional education approaches, AI-driven systems enable personalized learning experiences, predictive analytics for risk prevention, and interactive educational tools.

AI-powered educational platforms use machine learning algorithms to tailor content to individual patient needs. These platforms enhance engagement and comprehension through interactive modules, chatbots, and virtual tutors, improving rehabilitation adherence [13,14]. AI-based patient education in cardiac rehabilitation has been shown to improve psychological well-being and reduce anxiety[93] significantly. There are interactive patient education platforms to enhance engagement and knowledge retention. Studies show that AI-assisted interactive learning significantly improves patient comprehension and compliance with rehabilitation programs [94]. AI-enhanced translations also ensure accessibility for diverse populations by adapting educational materials for different languages and cultures [95]. Additionally, AI-based educational interventions have been found effective in preventing falls in stroke patients by improving their awareness and mobility strategies [96]. However, these studies are limited by small samples (e.g., $n = 35$ in [93]) and reporting biases.

3.5. Comparison between the traditional and proposed model

The traditional rehabilitation model (Fig. 1) relies on static, clinician-led cycles with limited adaptivity, often leading to inefficiencies like prolonged recovery times and lower patient engagement. In contrast, the novel AI-empowered model (Fig. 2) introduces real-time feedback loops, achieving efficiency gains such as a 20–30 % reduction in recovery time and a 25 % increase in patient adherence through adaptive interventions and immersive tools like virtual reality (VR) and LLMs [13,14]. A RCT study demonstrated that LLM-based cognitive games increased patient engagement up to 2 times in

traumatic brain injury rehabilitation [97]. The novel model was derived from literature gaps, including the lack of integrated, real-time systems and insufficient focus on patient-centered care in underserved populations [17,18]. The comparison according to several criteria is listed in Table 4. By addressing these gaps, the model enhances precision, accessibility, and scalability.

4. The challenges of transformation in clinical practices

The AI in rehabilitation science presents transformative potential, yet several challenges hinder its widespread clinical adoption. These barriers span technical, ethical, regulatory, and practical domains, necessitating multidisciplinary collaboration to address them effectively.

4.1. Data limitations and model generalizability

AI models typically require large, diverse datasets for training and validation. However, rehabilitation data often suffer from heterogeneity due to variations in patient demographics, impairment types, and rehabilitation protocols [17]. For instance, pediatric rehabilitation settings face unique challenges because datasets for children with disabilities are scarce, which limits the development of tailored AI applications [18]. Additionally, models trained in specific populations (e.g., adults with stroke) may fail to generalize to other groups (e.g., spinal cord injury patients), reducing their clinical utility. Specific solutions include multi-centre federated learning, as demonstrated in a study where federated learning for lesion segmentation in multiple sclerosis aggregated MRI data from multi-country centres, enhancing segmentation accuracy by 10 % while maintaining privacy [98].

4.2. Ethical and privacy concerns

The use of AI in rehabilitation raises ethical dilemmas, particularly regarding data privacy and algorithmic bias. Wearable sensors and tele-rehabilitation platforms collect sensitive physiological and movement data, necessitating robust encryption and compliance with regulations like GDPR and HIPAA [92,99]. Furthermore, biases in training data—such as underrepresentation of minority groups—may perpetuate disparities in care delivery. For example, AI-driven assessments of motor function might misinterpret movement patterns in patients from diverse cultural backgrounds, leading to inaccurate diagnoses [100]. Algorithmic interpretability is essential in clinical decisions to build trust and ensure accountability; Explainable AI (XAI) in rehabilitation fosters transparency and mitigates bias in multi-disease scenarios [47,64]. Transparent AI design and inclusive dataset curation are critical to mitigating these risks.

Table 4
Comparison of traditional and novel AI-empowered rehabilitation models.

Criterion	Traditional Model (Fig. 1)	Novel AI-Empowered Model (Fig. 2)
Efficiency	Static protocols, clinician-led; prolonged recovery (e.g., 6–12 months for stroke motor recovery)	Real-time adaptive interventions; reduction in recovery time
Patient Engagement	Limited interactivity; lower adherence (e.g., 50–60 % in home-based programs)	Immersive tools (VR, LLMs); higher adherence
Adaptivity	Manual adjustments at cycle starts; limited personalization	Dynamic adjustments via RL and CV
Accessibility	Clinic-based, limited access for rural/underserved populations	Tele-rehabilitation and home-based monitoring; reduction in hospital readmissions

4.3. Clinical integration and usability

Despite technological advancements, many AI tools remain confined to research settings due to poor integration with clinical workflows. Clinicians often lack the technical expertise to interpret AI outputs, while patients may struggle with complex interfaces, especially elderly populations [62]. For instance, exoskeletons controlled by RL algorithms require real-time adjustments that therapists must oversee, negating potential efficiency gains [69]. Challenges in integrating AI with existing healthcare systems include EHR data format compatibility, where mismatched standards (e.g., FHIR vs. proprietary formats) lead to interoperability issues. Solutions involve standardized APIs and clinician training programs. Simplifying user interfaces and providing clinician training are essential to fostering trust and adoption.

4.4. Validation and regulatory requirements

The transition from prototype to clinical application demands rigorous validation through randomized controlled trials (RCTs) and longitudinal studies. Many AI-driven interventions, such as digital twins for personalized therapy planning, lack evidence of long-term efficacy [31]. Regulatory frameworks for AI in healthcare also lag technological progress. For example, adaptive AI systems that evolve with patient progress may face scrutiny from agencies like the FDA, which traditionally evaluate static medical devices [48]. The importance of long-term follow-up needs to be highlighted. The long-term effects of AI technologies, as well as the attrition rates, have not been investigated adequately yet. Collaborative efforts between developers, clinicians, and regulators are needed to establish standardized evaluation protocols.

4.5. Cost-effectiveness considerations

Cost-effectiveness is a key consideration in the adoption of AI technologies for rehabilitation. While AI-driven interventions—such as robotic devices, wearable sensors, and virtual rehabilitation—can improve patient outcomes, personalize treatment, and increase accessibility, their high initial implementation costs remain a significant barrier to widespread use [101,102]. However, AI can also reduce long-term costs by enabling remote monitoring, decreasing workforce requirements, and potentially shortening recovery times and hospital stays, which may offset upfront expenses and improve overall healthcare efficiency [102–104].

Other cost-related challenges include technology literacy, user adherence, and data security, which can impact the successful integration and sustained use of AI in rehabilitation settings [101,102,104]. Studies highlight the need for robust clinical evaluations and real-world implementation research to better understand the true cost-effectiveness of these technologies, as current evidence is limited and sometimes inconsistent [104,105]. Future developments should focus on user-friendly, scalable, and ethically sound AI solutions to maximize both clinical and economic benefits in rehabilitation

5. Conclusion

AI has emerged as a transformative force in rehabilitation, enhancing

assessment accuracy, personalizing interventions, and extending access to remote and long-term care. AI-driven rehabilitation models integrate real-time monitoring, predictive analytics, and adaptive interventions, making rehabilitation more dynamic and patient-centered. However, several challenges must be overcome, including data heterogeneity, ethical concerns, regulatory complexities, and clinical adoption barriers. Future research should focus on improving AI model generalizability, ensuring ethical implementation, and fostering seamless integration into clinical workflows. As AI technology continues to evolve, its role in rehabilitation will expand, promising a future of more effective, inclusive, and accessible rehabilitation services.

Furthermore, AI holds immense potential to enhance rehabilitation accessibility in low- and middle-income countries (LMICs), aligning with the World Health Organization’s Rehabilitation 2030 initiative, which emphasizes universal access to high-quality rehabilitation services through strengthened governance, investment, and data systems, particularly in regions where up to 80 % of people with disabilities reside and services are limited. In LMICs, AI-driven solutions such as virtual rehabilitation platforms, wearable sensors for remote monitoring, and generative AI for personalized therapy can address workforce shortages and infrastructure gaps, enable home-based care and reduce healthcare burdens. By addressing existing challenges and leveraging AI’s capabilities, rehabilitation science can achieve unprecedented advancements in patient care and recovery.

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Availability of data and materials

The authors confirm that the data supporting the findings of this study are available within the article.

Funding

This work was supported by the National Natural Science Foundation of China (Grant No 82305357) and Fujian Provincial Natural Science Foundation (Grant No 2025J01948).

CRediT authorship contribution statement

Haoyang Liu: Writing – review & editing, Writing – original draft, Investigation, Conceptualization. **Qiuorong Xie:** Writing – review & editing, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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